PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : Surface Chemistry (Question Bank # 11)

MM : 100 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. The desorption of gas molecules from the adsorbent surface obeys Arrhenius equation. The average time upto which a N_2 molecule may remain adsorbed at Pt-surface at 400K is
[Given: Pre-exponential factor, $A = 1.25 \times 10^8 s^{-1}$;]
Activation energy of desorption = 16Kcal' $e^{20} = 5 \times 10^8$]
(A) 0.25 sec (B) 4 sec (C) 8 sec (D) 0.125 sec.

2. Smoke precipitator works on the principle of
(A) distribution law (B) neutralization of charge on colloids
(C) Le-Chaterlier's principle (D) addition of electrolytes

3. When $H_{2(gas)}$ gets adsorbed on transition metals hydride formation takes place. Which of the options will not be correct on the basis of above observation.
(A) Greater the surface area of transition metals greater will be the extent of adsorption.
(B) The enthalpy change involved will be in the range of 20-–40 kJ/mole.
(C) The extent of adsorption will increase with increase in temperature.
(D) The adsorption will be irreversible in nature.

4. Which of the following reaction can result in formation of a colloidal solution?
(A) Reduction of $AuCl_3$ with HCHO in presence of water.
(B) Reduction of $CO_{2(g)}$ using Mg to give carbon & MgO.
(C) Acid catalysed hydrolysis of ester.
(D) oxidation of $NH_{3(g)}$ into nitric oxide in Ostwald's process.

5. Which of the following is **not true** with respect to catalyst and catalytic action?
(A) A catalyst increases the rate of a reaction while inhibitor decreases the reaction rate.
(B) A promoter cannot catalyse the reaction on its own without the presence of catalyst.
(C) The activity of enzyme catalysts increases in presence of metal ions like Mn^{+2}, Co^{2+} etc.
(D) When a solid catalyst is used for gaseous reaction, energy released in physisorption increases the rate of reaction.

6. Which of the following process does not occur at the interface of phases?
(A) crystallisation (B) heterogenous catalysis
(C) homogeneous catalysis (D) corrosion

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7. Which ion has least coagulation value for +ve sol ?

(A) $[Fe(CN)_6]^{4-}$ (B) Cl^- (C) SO_4^{2-} (D) PO_4^{3-}

8. Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at 25ºC. For this process, the correct statement is -

(A) The adsorption requires activation at 25ºC

(B) The adsorption is accompanied by a decrease in enthalpy

(C) The adsorption increases with increase of temperature

(D) The adsorption is irreversible

9. A soap ($C_{17}H_{35}$ COONa) solution becomes a colloidal sol at a concentration of 1.2×10^{-3} M. On the average, 2.4×10^{13} micelle are present in 1 mm^3. What is the average number of stearate ions in one colloidal particle (micelle) : **Take : Avogadro's number = 6×10^{23}** (Assuming all the ions are present in form of micelle)

(A) 30 (B) 40 (C) 50 (D) 60

10. Promoters and Poison are the substance use in chemical reaction which

(A) Enhance and decrease the activity of catalyst respectively

(B) Decrease and enhance the activity of catalyst respectively

(C) Do not have any effect on catalyst

(D) Can be used in place of catalyst whenever required

(ii) One or more options correct Type

This section contains **11 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

11. Which of the following are true regarding micelles

(A) Micelles formation leads to net increase in entropy

(B) Hydrophobic portions are oriented within the cluster & the hydrophilic portions are exposed to the water phase

(C) After CMC no change in surface tension takes place, as the surface, becomes completely loaded with surfactant molecules and any further addition must arrange only as micelles.

(D) Micelles are not static, they are dynamic structures, so that the surfactant molecules leave the micelles & go into the solution, while other surfactant molecules enter the micelle from the solution.

12. Colloidal solution can be purified by

(A) Dialysis (B) Electrodialysis (C) Electrophoresis (D) Ultrafiltration

13. Which of the following is not lyophillic

(A) Gelatin sol (B) Silver sol (C) Sulphur sol (D) As_2S_3 sol

14. Colloidal Gold can be prepared by

(A) Bredig's are method (B) Reduction of $AuCl_3$

(C) Hydrolysis (D) Peptization

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15. The correct statement(s) about surface properties is (are) **[JEE 2017]**

(A) Cloud is an emulsion type of colloid in which liquid is dispersed phase and gas is dispersion medium

(B) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system.

(C) Brownian motion of colloidal particles does not depend on the size of the particles but depends on viscosity of the solution.

(D) The critical temperatures of ethane and nitrogen and 563 K and 126 K, respectively. The adsorption of ethane will be more than that of nitrogen on same amount of activated charcoal at a given temperature.

16. Choose the correct reason(s) for the stability of the **lyophobic** colloidal particle.

(A) Preferential adsorption of ions on their surface from the solution

(B) Preferential adsorption of solvent on their surface from the solution

(C) Attraction between different particles having opposite charges on their surface

(D) Potential difference between the fixed layer and the diffused layer of opposite charges around the colloidal particles

17. Tyndal effect is appriciable when

(A) The diameter of dispersed phase is comparable to the wavelength of light used.

(B) The diameter of dispersed phase is much smaller than the wavelength of light used

(C) The refractive indices of the dispersed phase and the dispersion medium must be same

(D) The refractive indices of the dispersed phase and the dispersion medium must differ greatly in magnitude.

18. Which of the following is/are true regarding given colloids :

(A) In sol solids dispersed in liquids

(B) In gels liquids dispersed in solids

(C) In emulsion liquid dispersed in liquids

(D) Smoke is an aerosol in which gas dispersed in solid

19. Which of the following is/are correct statement :

(A) Milk is an oil in water type emulsion

(B) Micelle formation takes place above a temperature called kraft temperature (T_k)

(C) Emulsifying agent are the component used to stabilize emulsion

(D) Protein and heavy metal salts are common emulsfying agent for O/W and W/O emulsions respectively

20. Which of the following statements are true about AgI sol prepared by peptization of AgI with $AgNO_3$.

(A) In electrophoresis, the dispered phase will move towards cathode.

(B) $AlCl_3$ is more effective than Na_3PO_4 for bringing about coagulation of sol.

(C) Na_3PO_4 is more effective than $AlCl_3$ for bringing about coagulation of sol.

(D) On persistant dialysis of sol, precipitation of coloidal sol take place (coagulation)

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21. Sodium palmitate is a major constituent in soap . When an aqueous solution of sodium palmitate having in 1kg of solvent is analysed for its boiling point, following data is obtained.

Moles	0	0.02	0.04	0.05	0.06	0.08
Boiling point	100°C	100.0372°C	100.0744°C	100.093°C	100.1°C	100.12°C

Identify the option(s) which is/are correct.

(A) Aqueous mixture of sodium palmitate having concentration equal to 0.07M will be heterogenous.

(B) The CMC value of sodium palmitate should be less than 0.06M.

(C) Osmotic pressure at 0.08 M at 300 K will be 1.9704 atm.

(D) At 0.4M concentration, the solution is homogenous.

(iii) Paragraph Type

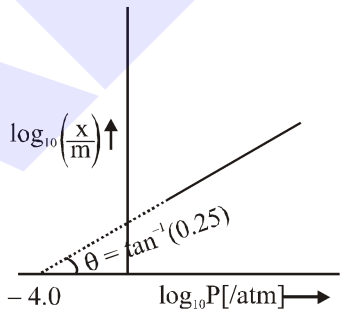
This section contains **1 paragraph** , describing theory, experiment, data etc. **2 questions** relate to one paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for question no. 22 & 23

The adsorption of a gas at a metal surface is called occlusion. The extent of gas adsorbed at metal surface at a certain temperature depends on pressure as

$$\frac{x}{m} = K.P^{\frac{1}{n}}$$

Where 'K' and 'n' are constant for a particular combination of gas and metal at fixed temperature and $\frac{x}{m}$ is the mass of gas adsorbed per unit mass of metal. For H_2 gas adsorbing at Pt-surface, the following graph is obtained.



22. The correct values of 'K' and 'n' are respectively.

(A) 1,0.25 (B) 10, 4

(C) 10, 0.25 (D) 0.10, 4.0

23. The mass of H_2 gas adsorbed on the surface of 10 gm Pt at 16 atm is

(A) 20 gm (B) 2 gm

(C) 4 gm (D) 200 gm

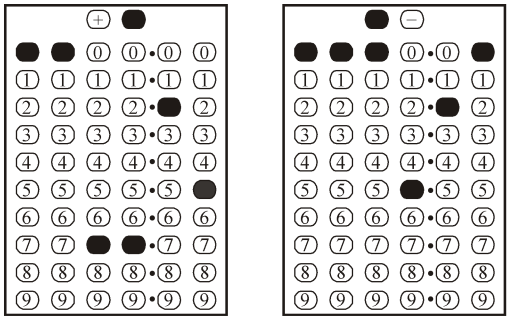
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SECTION-II: (Integer Second Decimal)

- This section contains **5** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

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1. Gold number of Haemoglobin is 0.030. Hence 100 ml of gold sol will require haemoglobin (in mg) so that gold is not coagulated by 10 ml of 10% NaCl solution.

2. A colloid prepared by the addition of KI to $AgNO_3$ solution is purified using dialysis. Find the minimum mass in grams of an electrolyte 'AB' (GMM = 60) required to completely coagulate 1ℓ of the aforementioned colloid

Given :

Active ion causing flocculation	Flocculation value(mmole/ℓ)
A^+	50
B^-	100

3. How many of the following statements are CORRECT -

(i) During adsorption, there is always a decrease in residual forces of the surface

(ii) During adsorption, there is decrease in surface energy which appears as heat.

(iii) Physisorption can be multilayered.

(iv) The ppt of $Mg(OH)_2$ attain blue color in presence of magneson, due to adsorption of magneson

(v) A mixture of noble gases can be separated by adsorption on coconut charcoal at different temperature.

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4. How many of the following phenomenon are observed due to process of coagulation.

(A) Delta formation at the meeting point of river and sea.

(B) Blue colour of sky.

(C) Precipitation of coal ash from smoke in chimney's.

(D) Destruction of lyophobic solution.

(E) Brownian motion.

(F) Cleaning action of detergents.

(G) Artificial rain.

(H) Use of alums in cleaning water.

5. Calculate the number of sols which are negatively charged.

$Fe_2O_3.xH_2O$ sol	Sb_2S_3 sol
Ag sol	Cu sol
Blood	basic dye
sol of clay	sol of starch
sol of sodium stearate	sol of sodium laurylsulphate
sol of charcoal.	

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ANSWER KEY

SECTION–I : (i) Only One option correct Type 3(–1)

1. Ans. (B)	2. Ans. (B)	3. Ans. (B)	4. Ans. (A)	5. Ans. (D)
6. Ans. (C)	7. Ans. (A)	8. Ans. (B)	9. Ans. (A)	10. Ans. (A)

(ii) One or more options correct Type 4(–1)

11. Ans. (A,B,C,D)	12. Ans. (A,B,D)	13. Ans. (B,C,D)	14. Ans. (A, B)
15. Ans. (B,D)	16. Ans. (A,D)	17. Ans. (A,D)	18. Ans. (A,B,C)
19. Ans. (A,B,C,D)	20. Ans. (A,C,D)	21. Ans. (A,B)	

(iii) Paragraph Type 3 (0)

22. Ans. (B)	23. Ans. (D)

SECTION-II: (Integer Second Decimal)

1. Ans. (0.30)	2. Ans. (6.00)	3. Ans. (5.00)	4. Ans. (5.00) A,C, D, G, H
5. Ans. (9.00)			

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